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• LS estimator:
$$\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = (X^T X)^{-1} X^T Y$$

Fitted values: $\hat{Y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h$

$$E(\hat{Y}_n) = \hat{\beta}_0 + \hat{\beta}_1 x_n = (1 \ x_n) \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}$$

$$\begin{aligned} \text{Var}(\hat{Y}_n) &= \text{Var}(\hat{\beta}_0) + x_n^2 \text{Var}(\hat{\beta}_1) + 2x_n \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &= (1 \ x_n) \text{Var}(\underline{\hat{\beta}}) \begin{pmatrix} 1 \\ x_n \end{pmatrix} \end{aligned}$$

$$\text{Var}(\underline{\hat{\beta}}) = \sigma^2 (X^T X)^{-1}$$

$$\begin{aligned} \hat{Y}_n &= \hat{\beta}_0 + \hat{\beta}_1 x_n = (\hat{\beta}_0 + \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 (x_n - \bar{x}) \\ &= \bar{y} + \hat{\beta}_1 (x_n - \bar{x}) \end{aligned}$$

$$= \sigma^2 \left(\frac{1}{n} + \frac{(x_n - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right) \quad (\# \text{ Now})$$

Recall: $\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right)$

* Characterization of conditional expectation:

$E(Y|X)$ minimizes $E[(Y - g(X))^2]$ among all possible $g(X)$ s.t. $E[(g(X))^2] < \infty$

- Predicted value of Y for given $X = x_n$ is also $\hat{Y}_n$

$$Y_{\text{new}} = \beta_0 + \beta_1 x_n + \epsilon_{\text{new}}$$

- Mean squared prediction error = $E[(Y_{\text{new}} - \hat{Y}_h)^2]$

$$= E[(\epsilon_{\text{new}} + \{(\beta_0 + \beta_1 x_n) - (\hat{\beta}_0 + \hat{\beta}_1 x_n)\})^2]$$

Independent

$$= \sigma^2 + \text{Var}(\hat{Y}_h) = \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_n - \bar{x})^2}{S_x^2} \right]$$

⇒ Test for regression coeff.:

Test for lack of dependence on $X \equiv \text{Test } H_0: \beta_1 = 0$

" " no intercept $\equiv \text{Test } H_0: \beta_0 = 0$

More generally, we may test: $H_0: \beta_1 = \bar{\beta}_1$
 $H_0: \beta_0 = \bar{\beta}_0$ } specified

Even more generally: $H_0^{(1)}: \beta_0 + 2\beta_1 = 5 \Leftrightarrow H_0: \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 5 \end{pmatrix}$
 $H_0^{(2)}: \beta_0 + \beta_1 = 3 \Leftrightarrow H_0: \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 3 \end{pmatrix}$
 $\Leftrightarrow H_0: \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

• Test for individual parameter:

$H_0: \beta_0 = \bar{\beta}_0$ or $H_0: \beta_1 = \bar{\beta}_1$

Typical test: $H_0: \theta = \bar{\theta}$ v/s $H_1: \theta \neq \bar{\theta}$

Ideally $\left(\frac{\hat{\theta} - \bar{\theta}}{s(\hat{\theta})} \right)$ should be free of θ at least.

under H_0 . $\hookrightarrow$ This is called a pivotal qtt.

— Test: $H_0: \beta_0 = \bar{\beta}_0$ v/s $H_1: \beta_0 \neq \bar{\beta}_0$

$$T_0 = \frac{\hat{\beta}_0 - \bar{\beta}_0}{s(\hat{\beta}_0)}, \text{ where } s(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_x^2} \right)}$$

$$\hat{\sigma}^2 = \text{MSE}$$

$$= \frac{\text{SSE}}{n-2} = \frac{\sum (Y_i - \bar{Y})^2}{n-2}$$

(★) Assume $Y_i \stackrel{\text{indep}}{\sim} N(\beta_0 + \beta_1 X_i, \sigma^2)$

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_x^2}\right)\right)$$

$$\Rightarrow \frac{\hat{\beta}_0 - \beta_0}{\sigma \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{S_x^2}}} \sim N(0, 1)$$

Let this be V_0

Then, $T_0 = \frac{V_0}{\sqrt{\hat{\sigma}^2 / \sigma^2}}$

We know, $\frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-2}^2$

Also, $\hat{\beta}_0$, $\hat{\beta}_1$ & SSE are indep.

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = (X^T X)^{-1} X^T Y = \beta + (X^T X)^{-1} X^T \varepsilon$$

$$SSE = Y^T (I - H) Y = \|(I - H) Y\|^2 \quad (H = X(X^T X)^{-1} X^T)$$

Check that $(I - H)\varepsilon$ & $\hat{\beta}$ are uncorrelated

Hint: $E[(I - H)\varepsilon \varepsilon^T X (X^T X)^{-1}] = \sigma^2 \underbrace{(I - H) X (X^T X)^{-1}}_{=0}$

$$\therefore T_0 = \frac{V_0}{\sqrt{\hat{\sigma}^2 / \sigma^2}} \stackrel{H_0}{\sim} \frac{N(0, 1)}{\sqrt{\frac{\chi_{n-2}^2}{n-2}}} \quad \text{indep.}$$

$$\therefore T_0 \sim t_{n-2}$$

$\therefore$ Test statistic: $T_0 = \frac{\hat{\beta}_0 - \bar{\beta}_0}{s(\hat{\beta}_0)}$ Under H_0 t_{n-2}

Reject H_0 in favour of H_1 if $|T_0| > t_{n-2}^{1-\alpha/2}$

- Now test: $H_0: c_0\beta_0 + c_1\beta_1 = v$ v/s $H_1: c_0\beta_0 + c_1\beta_1 \neq v$

Test statistic $\frac{(c_0\hat{\beta}_0 + c_1\hat{\beta}_1) - v}{s(c_0\hat{\beta}_0 + c_1\hat{\beta}_1)}$ Under $H_0 \sim t_{n-2}$

$$\text{Var}(c_0\hat{\beta}_0 + c_1\hat{\beta}_1) = \sigma^2 (c_0 \ c_1)(X^T X)^{-1} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$$

Q & $s(c_0\hat{\beta}_0 + c_1\hat{\beta}_1) = \sqrt{\text{Var}(c_0\hat{\beta}_0 + c_1\hat{\beta}_1)}$

- Test for $H_0: \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} \bar{\beta}_0 \\ \bar{\beta}_1 \end{pmatrix}$ v/s $H_1: \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \neq \begin{pmatrix} \bar{\beta}_0 \\ \bar{\beta}_1 \end{pmatrix}$

Suggestion - Bonferroni method for simultaneous testing:

$$A_0 = \left\{ \left| \frac{\hat{\beta}_0 - \bar{\beta}_0}{s(\hat{\beta}_0)} \right| > t_{n-2}^{1-\alpha_0/2} \right\} \quad P_{H_0}(A_0) = \alpha_0$$

$$A_1 = \left\{ \left| \frac{\hat{\beta}_1 - \bar{\beta}_1}{s(\hat{\beta}_1)} \right| > t_{n-2}^{1-\alpha_1/2} \right\} \quad P_{H_0}(A_1) = \alpha_1$$

α_0 & α_1 are such that: $\alpha_1 + \alpha_0 \leq \alpha$

$$P_{H_0}(A_0 \cup A_1) \leq \alpha$$

If $\alpha_0 = \alpha_1 = \frac{\alpha}{2}$, it's Bonferroni method.

- $\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}$, $\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$

Under $H_0: \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} - \begin{pmatrix} \bar{\beta}_0 \\ \bar{\beta}_1 \end{pmatrix} \sim N(0, \sigma^2 (X^T X)^{-1})$

$$\left(\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} - \begin{pmatrix} \bar{\beta}_0 \\ \bar{\beta}_1 \end{pmatrix} \right)^T (s^2(\hat{\beta}))^{-1} \left(\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} - \begin{pmatrix} \bar{\beta}_0 \\ \bar{\beta}_1 \end{pmatrix} \right)$$

$$= (\hat{\beta} - \bar{\beta})^T (\hat{\sigma}^2 (X^T X)^{-1})^{-1} (\hat{\beta} - \bar{\beta})$$

$$= \frac{(\hat{\beta} - \bar{\beta})^T (\sigma^2 (X^T X)^{-1})^{-1} (\hat{\beta} - \bar{\beta})}{\hat{\sigma}^2 / \sigma^2} \underset{H_0}{\sim} \chi^2_2$$

$$\therefore T_0 = (\hat{\beta} - \bar{\beta})^T (\hat{\sigma}^2 (\hat{\beta}))^{-1} (\hat{\beta} - \bar{\beta}) \underset{H_0}{\sim} 2 F_{2, n-2}$$